

Fine-Tuned DistilBERT for IMDb Sentiment Classification Baseline Comparison and Probability Calibration

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Abstract—Transformer models are widely used for sentiment classification, but their practical advantage over strong linear baselines and the reliability of their confidence estimates are often underexamined. This study presents a reproducible CUDA-based evaluation of a fine-tuned DistilBERT classifier on the IMDb Dataset of 50K Movie Reviews. After removing 418 duplicate reviews, 49,582 unique samples were divided using a stratified 80/10/10 train-validation-test split. DistilBERT was fine-tuned for two epochs and compared with a word-unigram and bigram TF-IDF logistic-regression baseline tuned only on the validation set. On 4,959 held-out reviews, DistilBERT achieved 91.51% accuracy, an F1-score of 0.9155, ROC-AUC of 0.9724, and average precision of 0.9705. The baseline achieved 91.45% accuracy, the same F1-score, ROC-AUC of 0.9715, and the same average precision. An exact paired McNemar test found no evidence of an accuracy difference ($p=0.9258$). Temperature scaling fitted exclusively on validation logits reduced test negative log-likelihood from 0.3011 to 0.2275 and expected calibration error from 0.0615 to 0.0321 without changing classification accuracy. The results show that calibration materially improved probability reliability, whereas model-complexity gains over a strong sparse baseline were negligible on this balanced benchmark.

Keywords—sentiment analysis, DistilBERT, IMDb, text classification, TF-IDF, probability calibration, temperature scaling, CUDA

I. INTRODUCTION

Sentiment analysis assigns affective labels to text and remains a standard test bed for natural language processing (NLP). Movie-review classification is particularly useful because documents are long enough to contain compositional phenomena such as negation, contrast, and mixed opinion, while labels are simple enough to support controlled comparison. The Large Movie Review Dataset introduced by Maas et al. contains 25,000 labeled training reviews and 25,000 labeled test reviews and has become a widely used binary-sentiment benchmark [1]. The Kaggle distribution used in this study combines these reviews into a single 50,000-row comma-separated file [2].

Pretrained Transformer encoders changed text classification by replacing fixed sparse representations with contextual token representations. The Transformer architecture uses self-attention to model token interactions without recurrence [3], and BERT demonstrated that a pretrained bidirectional encoder can be adapted to downstream tasks by fine-tuning a small task-specific head [4]. DistilBERT compresses BERT through knowledge distillation and reduces computational demand while retaining much of its language-understanding performance [5]. These properties make DistilBERT attractive for local fine-tuning on a consumer NVIDIA GPU.

High predictive accuracy alone, however, does not establish that a complex model is operationally preferable. A TF-IDF representation paired with logistic regression is

inexpensive, interpretable at the feature level, and often highly competitive on topic and sentiment benchmarks. In addition, neural classifiers may produce overconfident softmax probabilities. If probability values are shown to users or used to trigger downstream actions, their agreement with observed correctness matters independently of accuracy. Temperature scaling offers a simple post-hoc correction by fitting a single positive parameter on held-out data [6].

This study addresses three practical questions. First, how well does a CUDA-trained DistilBERT classifier perform on a strictly held-out IMDb test partition after duplicate removal? Second, does the transformer materially outperform a tuned TF-IDF logistic-regression baseline under the same split? Third, can validation-fitted temperature scaling improve probability calibration without changing discrimination? The contribution is an end-to-end, reproducible evaluation that combines a neural model, a strong classical baseline, error analysis, calibration, and an interactive inference application. No new neural architecture is proposed; the value lies in controlled comparison and transparent reporting.

II. RELATED WORK

A. Sentiment Classification

Early sentiment systems commonly used bag-of-words or n-gram features with linear classifiers. TF-IDF weighting emphasizes terms that are frequent within a document but less common across the collection; comparative analyses of term-weighting schemes established its effectiveness for information retrieval and text representation [7]. Logistic regression then provides a linear decision function and probabilistic output over the sparse feature vector. Such methods remain strong when lexical polarity is directly associated with the target label.

The IMDb dataset was introduced with learned representations that combined unsupervised and supervised signals [1]. Later work increasingly relied on pretrained contextual encoders. BERT jointly conditions on left and right context and can be fine-tuned with a task-specific output layer [4]. DistilBERT transfers knowledge from a larger BERT teacher during pretraining, yielding a smaller six-layer model intended to reduce inference cost [5]. The Hugging Face Transformers library provides consistent implementations and pretrained checkpoints for these architectures [8].

B. Confidence Calibration

A classifier is calibrated when predictions assigned confidence q are correct approximately q of the time. Earlier comparative work established that probability quality can differ substantially even when classifiers have similar accuracy [12]. Guo et al. showed that modern neural networks can be accurate yet systematically miscalibrated and found

temperature scaling to be an effective post-hoc method across several settings [6]. Calibration is distinct from discrimination: multiplying logits by the inverse of a positive temperature preserves their ordering and therefore does not change the argmax class. It can nevertheless improve negative log-likelihood (NLL), Brier score, and expected calibration error (ECE).

III. MATERIALS AND METHODS

A. Dataset and Preprocessing

The experiment used the Kaggle release titled IMDB Dataset of 50K Movie Reviews [2]. Each record contains a review and a binary sentiment label. HTML break tags were replaced with spaces, repeated whitespace was collapsed, and leading and trailing whitespace was removed. Records with missing required fields were excluded. Sentiment strings were normalized to lower case and mapped to 0 for negative and 1 for positive.

Duplicate removal occurred before splitting to prevent an identical review from appearing in more than one partition. Of the 50,000 loaded records, 418 duplicates were removed, leaving 49,582 unique reviews. A fixed random seed of 42 and stratification by sentiment were used to construct an 80% training partition, a 10% validation partition, and a 10% test partition. The test set was not used for model or hyperparameter selection.

TABLE I
DATA PARTITIONS AFTER DUPLICATE REMOVAL

Partition	Reviews	Share
Training	39,665	80.0%
Validation	4,958	10.0%
Test	4,959	10.0%
Total	49,582	100%

B. DistilBERT Fine-Tuning

The pretrained distilbert-base-uncased checkpoint was loaded with a newly initialized binary sequence-classification head. Reviews were tokenized with the corresponding uncased WordPiece tokenizer and truncated to 256 tokens. Dynamic padding was applied within each mini-batch. Optimization used AdamW with a learning rate of 2×10^{-5} , weight decay of 0.01, mixed-precision FP16 computation, and two training epochs. The per-device batch size was 8, with two-step gradient accumulation for an effective batch size of 16. Validation and checkpointing occurred every 500 update steps, and the checkpoint with the highest validation F1-score was restored. The run used one NVIDIA GeForce RTX 4060 Laptop GPU with 8 GB of video memory.

TABLE II
DISTILBERT TRAINING CONFIGURATION

Parameter	Value
Checkpoint	distilbert-base-uncased
Maximum length	256 tokens
Epochs	2
Batch size	8
Gradient accumulation	2 steps
Effective batch	16
Learning rate	2×10^{-5}
Weight decay	0.01
Precision	FP16
Selection metric	Validation F1

C. TF-IDF Logistic-Regression Baseline

The baseline used word unigrams and bigrams with lowercase conversion, Unicode accent stripping, sublinear term frequency, a minimum document frequency of 2, a maximum document frequency of 0.98, and a vocabulary capped at 100,000 features. The vectorizer was fitted only on the training partition. Logistic regression used the liblinear solver with a maximum of 1,000 iterations. The inverse regularization strength C was selected from {0.5, 1, 2, 4} by validation F1-score; C=4 was selected. The fitted vectorizer and classifier were then evaluated once on the unchanged test partition. Scikit-learn supplied the baseline and evaluation routines [9].

D. Evaluation Measures

Primary classification measures were accuracy, positive-class precision, positive-class recall, and F1-score at a probability threshold of 0.5. Threshold-independent discrimination was measured by receiver operating characteristic area under the curve (ROC-AUC) and average precision (AP). The confusion matrix was reported to expose the direction of errors. These complementary measures avoid relying on accuracy alone [10].

Because both classifiers were evaluated on the same reviews, their binary correctness outcomes were compared with a two-sided exact McNemar test [13]. This paired test conditions on reviews for which the models disagreed in correctness and tests whether the two discordant counts are equally likely. Statistical significance was assessed at the 0.05 level.

Calibration was evaluated with NLL, multiclass Brier score, and ECE using 15 equal-width confidence bins. Temperature scaling divided each logit by a scalar temperature T before the softmax operation. T was optimized on validation logits by minimizing NLL and was then fixed for test evaluation. This separation prevents the test set from influencing the calibration parameter.

E. Implementation and Reproducibility

Training and neural inference used PyTorch [11] and Hugging Face Transformers [8]. The implementation explicitly checked CUDA availability and terminated rather than silently falling back to CPU. Random seeds governed the

split and model-training data order. The complete workflow generated saved model weights, tokenization artifacts, metrics in JSON and CSV formats, test-level predictions, misclassified examples, diagnostic plots, the calibrated temperature, and a Streamlit inference interface. The training run required 919.1 s (approximately 15.3 min), processing 86.31 samples/s on the stated GPU.

IV. RESULTS

A. Held-Out Classification Performance

On 4,959 held-out reviews, DistilBERT achieved an accuracy of 0.9151, precision of 0.9143, recall of 0.9168, and F1-score of 0.9155. ROC-AUC was 0.9724 and AP was 0.9705. The tuned TF-IDF logistic-regression model achieved 0.9145 accuracy, 0.9083 precision, 0.9229 recall, 0.9155 F1, 0.9715 ROC-AUC, and 0.9705 AP.

TABLE III
HELD-OUT TEST PERFORMANCE

Model	Accuracy	F1	ROC-AUC	AP
DistilBERT	0.9151	0.9155	0.9724	0.9705
TF-IDF + LR	0.9145	0.9155	0.9715	0.9705

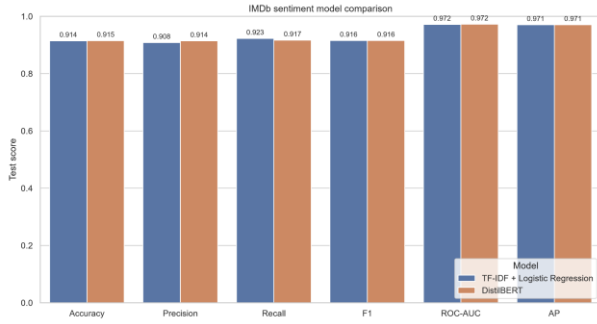


Fig. 1. Held-out test performance of the tuned TF-IDF logistic-regression baseline and fine-tuned DistilBERT.

The absolute accuracy difference was 0.0006, equivalent to 0.06 percentage points. F1 and AP were identical to four decimal places, and the ROC-AUC difference was 0.0009. Paired predictions agreed in correctness on 4,498 reviews: both models were correct on 4,306 and both were wrong on 192. Among the 461 discordant cases, DistilBERT alone was correct on 232 and the baseline alone on 229. A two-sided exact McNemar test yielded $p=0.9258$, providing no evidence that their accuracies differed on this test partition.

B. Error Distribution

The DistilBERT confusion matrix contained 2,256 true negatives, 214 false positives, 207 false negatives, and 2,282 true positives (Fig. 2). The baseline contained 2,238 true negatives, 232 false positives, 192 false negatives, and 2,297 true positives. Thus, relative to the baseline, DistilBERT removed 18 false positives but added 15 false negatives, matching its higher precision and lower recall. DistilBERT negative-class precision, recall, and F1 were approximately 0.916, 0.913, and 0.915; positive-class values were approximately 0.914, 0.917, and 0.916.

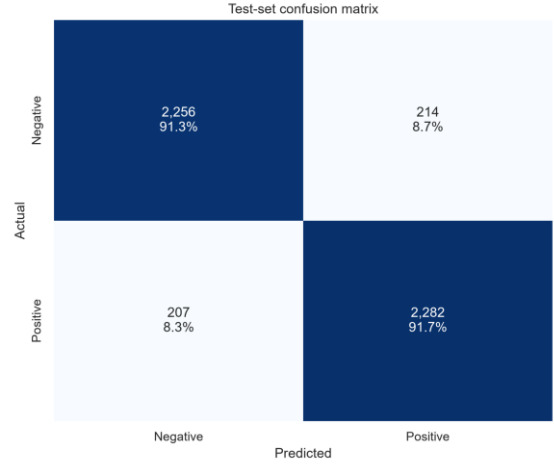


Fig. 2. Confusion matrix for the held-out DistilBERT test predictions. Percentages are normalized within each actual class.

C. Ranking Performance

The ROC curve remained well above chance across operating points, producing an AUC of 0.9724. The precision-recall curve also remained high over most recall values, with AP of 0.9705 (Fig. 3). These threshold-independent results show that the model ranked positive reviews effectively even though a fixed threshold still misclassified 421 reviews.

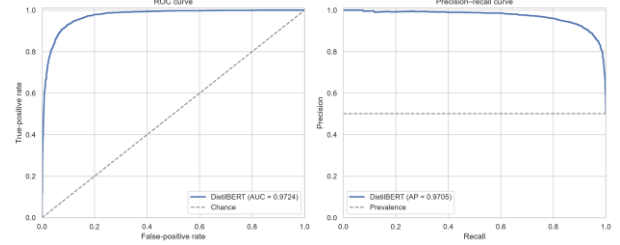


Fig. 3. Receiver operating characteristic and precision-recall curves for DistilBERT on the test partition.

D. Training Dynamics

Training loss declined from above 1.0 early in optimization to roughly 0.3 near the end of the second epoch. Validation accuracy increased from 0.8810 at the first evaluation to 0.9221 at the final evaluation, while validation F1 increased from 0.8884 to 0.9224. Validation loss reached its lowest observed value of approximately 0.234 early in training and subsequently increased to approximately 0.289 despite continued improvement in classification metrics (Fig. 4). This divergence indicates increasing confidence and mild overfitting in probabilistic terms, motivating post-hoc calibration.

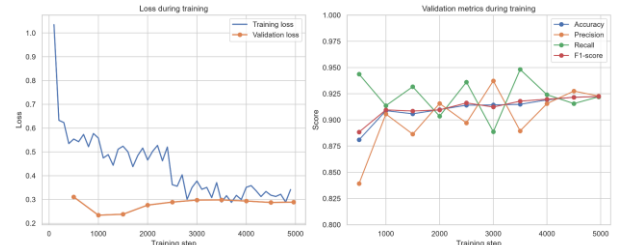


Fig. 4. Training loss and validation metrics across optimization steps. Validation loss rises after its early minimum while F1 continues to improve.

E. Probability Calibration

Validation optimization produced a temperature of 1.857998. A temperature greater than one softens the logits, which is consistent with an overconfident uncalibrated model. On the untouched test set, temperature scaling reduced NLL from 0.301062 to 0.227487, Brier score from 0.071906 to 0.065014, and ECE from 0.061494 to 0.032140. The relative reductions were 24.4%, 9.6%, and 47.7%, respectively. Accuracy remained 0.9151 and F1 remained 0.9155 because positive temperature scaling does not alter the predicted class.

TABLE IV
TEST CALIBRATION BEFORE AND AFTER SCALING

Metric	Before	After	Reduction
NLL	0.3011	0.2275	24.4%
Brier score	0.0719	0.0650	9.6%
ECE	0.0615	0.0321	47.7%

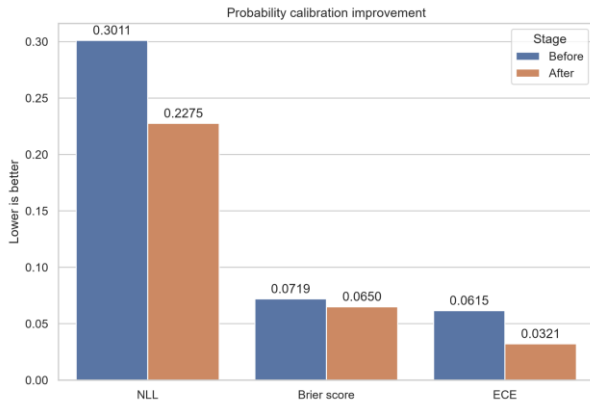


Fig. 5. Lower test calibration error after validation-fitted temperature scaling.

V. DISCUSSION

A. Transformer Versus Sparse Baseline

The central empirical result is not that DistilBERT decisively beat the baseline; it did not. The transformer improved accuracy by only 0.06 percentage points and ROC-AUC by 0.0009, while F1 and AP were unchanged at the reported precision. The paired McNemar result confirms that the three-review difference among discordant outcomes is compatible with chance. For this balanced and lexically explicit benchmark, word unigrams and bigrams capture much of the signal needed for classification. The baseline therefore offers a compelling efficiency-performance tradeoff when memory, latency, interpretability, or deployment simplicity matters.

DistilBERT still offers capabilities that are not reflected fully by aggregate IMDb metrics. Contextual encoders can model token interactions, transfer to related tasks, and support continued fine-tuning. However, such general advantages should not be confused with demonstrated superiority in the present experiment. A deployment decision should compare latency, energy use, memory, maintenance burden, and robustness under domain shift, not select the transformer merely because it is more modern.

B. Why Calibration Was the More Meaningful Gain

The calibration experiment produced a clearer improvement than the model comparison. The unscaled network assigned very high confidence to many predictions, including errors. Temperature scaling reduced all three proper or calibration-sensitive measures while preserving discrimination and class decisions. This matters in an interactive application because a displayed 99% probability can imply a degree of certainty unsupported by observed outcomes. The calibrated probabilities are still estimates, but they are less extreme and better aligned with empirical correctness.

Calibration must be fitted on data separate from final testing. Optimizing temperature on the test partition would leak outcome information and bias reported improvement. The present workflow instead fitted one scalar on the validation partition and evaluated it once on the held-out test partition. This design is simple and reproducible, although calibration should be reassessed after domain or label-distribution shifts.

C. Error Sources and Operational Interpretation

Inspection of sentiment errors is important because binary movie reviews can contain sarcasm, conflicting clauses, plot summary mixed with evaluation, quoted dialogue, or a favorable comment embedded in an otherwise negative review. Truncation at 256 tokens can also omit a decisive conclusion near the end of a long review. The roughly symmetric false-positive and false-negative counts indicate no severe directional bias at the 0.5 threshold, but aggregate balance does not reveal linguistic subgroups or review-length effects.

The Streamlit interface demonstrates local CUDA inference and reports calibrated probabilities. It should be interpreted as a research demonstration rather than an unrestricted sentiment service. User-entered text outside English movie reviews represents distribution shift. Probability calibration on IMDb does not guarantee calibration for product reviews, social media, clinical text, or other domains.

VI. LIMITATIONS AND THREATS TO VALIDITY

This study has several limitations. First, it uses one dataset and one random split. Repeated stratified splits or cross-validation would provide a distribution of performance differences, although full transformer retraining would increase computation. Second, only one transformer checkpoint and a limited set of hyperparameters were evaluated. The study therefore cannot generalize to all pretrained language models or establish the optimal DistilBERT configuration.

Third, although the paired test found no detectable accuracy difference on this partition, it does not establish model equivalence and does not account for variability from retraining or resplitting. Reviews were also truncated to 256 tokens, which improves feasibility on an 8 GB GPU but may remove sentiment-bearing context. Fourth, no external dataset was used to measure robustness under domain, temporal, or source shift.

Fifth, temperature scaling corrects average confidence but does not guarantee calibration within subgroups or all confidence intervals. ECE is sensitive to binning, and the 15-bin estimate should be interpreted alongside NLL and Brier score. Finally, duplicate removal was based on the available review text; near-duplicate reviews were not detected. Future work should use semantic near-duplicate checks, multiple seeds, external validation, calibration confidence intervals, latency and energy benchmarks, and systematic error annotation.

VII. CONCLUSION

A fine-tuned DistilBERT classifier achieved strong held-out IMDb performance on a consumer CUDA GPU, reaching 91.51% accuracy, 0.9155 F1, and 0.9724 ROC-AUC. Nevertheless, a tuned TF-IDF logistic-regression baseline reached nearly identical performance, and a paired exact McNemar test found no significant accuracy difference. In contrast, validation-fitted temperature scaling substantially improved NLL, Brier score, and ECE without changing accuracy. The study supports two practical conclusions: strong classical baselines should accompany transformer evaluations, and calibrated confidence should be reported when probabilities are exposed to users. Future experiments should prioritize external validation, multiple training seeds, and efficiency measurements before claiming a deployment advantage for the transformer.

DATA AVAILABILITY

The IMDb Dataset of 50K Movie Reviews used in this study is publicly available through Kaggle [2]. The original Large Movie Review Dataset and its accompanying publication are described by Maas et al. [1].

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